

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The network protocol analyzer logs indicate port 53 is unreachable while requesting the yummyrecipe website. *Port 53 is the standard network port used by the Domain Name System (DNS) to translate human-readable website names into numeric IP addresses. This may indicate a problem with the DNS server or the firewall configuration. It is possible that this is an indication of a malicious attack on the DNS server.*

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:

The incident occurred at approximately **13:24 (1:24 p.m.)**, as indicated by the tcpdump timestamp **13:24:32.192571**.

Explain how the IT team became aware of the incident:

The IT team became aware of the incident after users reported that they were unable to access **yummyrecipesforme.com** and received a “**destination port unreachable**” error when attempting to visit the website.

Explain the actions taken by the IT department to investigate the incident:

The IT team attempted to access the website to reproduce the problem. They then used the network protocol analyzer **tcpdump** to capture and examine network traffic while trying to load the webpage. The team analyzed the DNS requests sent from the user's computer and the responses returned by the DNS server.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

The investigation revealed that the user's computer, with IP address

192.51.100.15, sent DNS requests using **UDP** to the DNS server at **203.0.113.2**. The DNS requests were directed to **UDP port 53**, which is the standard port used for DNS services. Instead of receiving a DNS response containing the website's IP address, the computer received **ICMP error messages stating "udp port 53 unreachable."** The same error occurred multiple times. Because the DNS request could not be completed, the browser was unable to resolve the website's domain name to an IP address and could not continue with the HTTPS connection.

Note a likely cause of the incident:

A likely cause of the incident is that the **DNS service on UDP port 53 was unavailable or not listening for requests**. Another possibility is that a firewall or network configuration was blocking UDP traffic to port 53. Further investigation of the DNS server, firewall rules, and DNS service configuration would be required to determine the exact root cause.